

Cell Differentiation Task

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1. Idea

The idea was to devise a toy problem of the sort where each individual cell does not have a pre-defined target. The property trained for should be measured at the grid level, which would subsequently mean that loss would also be applied at grid level.

Such a problem can bring out the difference between a Moore-neighbourhood based NCA and a NCA with non-local connections better than problems with cell level targets. Using such a problem, we can test different versions of the non-local NCA and treat the advantage over vanilla NCA as a “Hello World” check.

The problem is defined as follows:

1. Each grid starts out with each cell in a parent state.
2. A parent state can mutate into a daughter state A, or daughter state B, or stay as it is.
3. The goal is to learn a set of update rules, so that the NCA settles into a configuration, where the ratio of daughter A to daughter B is a predefined target, and the number of cells in the parent state is as minimum as possible.

It is entirely possible for the vanilla NCA to solve this task. The advantage discussed earlier would be in the form of convergence time, since immediate neighbour-neighbour diffusion would take a longer time than non-local version.

- Since if a cell has a predefined target, it can reach there without little consideration given towards the state of the grid.

- where information can travel faster; if it is a small-world map, information can travel within $\log(N)$ steps where N is the grid size.

2. Design

The vanilla NCA architecture is pretty much the same as that of the one discussed in A. Mordvintsev, E. Randazzo, E. Niklasson, and M. Levin [1]. Main differences are as follows:

- We will have no hidden channels for the first few experiments. We will observe the role hidden channels play by varying them after initial experiments are performed. Similarly, the firing rate of cells is kept at 1, and will be varied later.
- The Sample Pool structure is kept as is from [1]. The worst performing grid is replaced with a grid at initial state after every epoch

Some task specific design choices:

1. The main channel is responsible for the **fate** of the cell.

$$\text{fate}(f) = \begin{cases} \text{parent} , & -0.95 \leq f \leq 0.95 \\ \text{daughter A} , & -1 \leq f < -0.95 \\ \text{daughter B} , & 0.95 < f \leq 1 \end{cases} \quad (1)$$

The update output from the network is capped at $\text{mod}(0.05)$ and the cell fate is capped at $\text{mod}(1)$.

2. Loss functions

1. **Commitment Loss:** a double wellled potential with two minima, one at -1 and the other at 1 , and a maximum at 0 . This forces the cells to move away from the parent state towards the two daughter states.

$$\mathcal{L}_{\text{commitment}} = (1 - \text{fate}^2)^2 \quad (2)$$

2. **Ratio Loss:** quantifies the ratio of the grid from the target ratio. This is done with the help of a helper function called strength.

$$\begin{aligned} \text{strength}(A) &= \sum_{\text{grid}} \text{ReLU}(\text{fate}) \\ \text{strength}(B) &= \sum_{\text{grid}} \text{ReLU}(-\text{fate}) \end{aligned} \quad (3)$$

Then ratio of B is defined as

$$\text{ratio}_B = \frac{\text{strength}(B)}{\text{strength}(A) + \text{strength}(B)} \quad (4)$$

The ratio loss is then

$$\mathcal{L}_{\text{ratio}} = E_{\text{batch}} [(\text{ratio}_B - \text{target ratio})^2] \quad (5)$$

• with Moore neighbours

• strength quantifies the commitment towards a daughter

• The ratio loss is computed across the batch used during training epoch.

3. **Variance Loss** : If the batch size gets large enough, the network can push each individual grid within the batch towards all daughter cell A or all daughter cell B and try to achieve the ratio at the batch level instead of the grid level. In order to mitigate this we try to minimize the variance between grids.

$$\mathcal{L}_{\text{variance}} = E_{\text{batch}} \left[(\text{ratio}_B - E_{\text{batch}}[\text{ratio}_B])^2 \right] \quad (6)$$

The total loss is

$$\mathcal{L} = \mathcal{W}_{\text{commitment}} * \mathcal{L}_{\text{commitment}} + \mathcal{W}_{\text{ratio}} * \mathcal{L}_{\text{ratio}} + \mathcal{W}_{\text{variance}} * \mathcal{L}_{\text{variance}} \quad (7)$$

where $\mathcal{W}$ s are the weights to the terms.

2.1. Vanilla NCA

The pipeline is given below:

1. An identity filter, Sobel X filter, and Sobel Y filter are convolved over the grid ($C \rightarrow 3C$)
2. A CNN with 3 Layers act on this grid with $3C$ channels:

Layer 1	2D Convolution , 32 Channels, ReLU, Width 1	$3C \rightarrow 32$
Layer 2	2D Convolution , 32 Channels , ReLU, Width 1	$32 \rightarrow 32$
Layer 3	2D Convolution , C Channels , No Activation, Width 1	$32 \rightarrow C$

3. The output of Layer 3 is the update given to the current grid.
4. The grid is evolved for some steps and the loss is computed. The gradients are updated via BPTT.
5. The worst performing grid is removed and replaced with a grid in the initial state
6. Steps 1 to 5 are repeated for some number of times.

• C is the number of channels

• who has the highest variance in the batch

2.2. Non Local NCA

Non locality is established by randomly selecting a fixed percentage of cells in the grid to have one long range partner. The partner is also randomly chosen, and their distance must be greater than some fixed Manhattan Distance . A wiring bank is made by repeating the long range wiring process for N_{wirings} times. When a batch is selected from the pool during train, each sample within the batch is chosen to be a unique wiring from the wiring bank.

• here it is 10

• since it is random, each wiring process generates a different graph

• this wirings associated with a batch change in each epoch

This ensures that the network learns rules which generalize to all kinds of wirings..

2.2.1. v1 (Discarded)

1. An identity filter, Sobel X filter, and Sobel Y filter are convolved over the grid ($C \rightarrow 3C$)
2. A CNN (termed Local CNN) with 3 Layers act on this grid with $3C$ channels:

Layer 1	2D Convolution , 32 Channels, ReLU, Width 1	$3C \rightarrow 32$
Layer 2	2D Convolution , 32 Channels , ReLU, Width 1	$32 \rightarrow 32$
Layer 3	2D Convolution , C Channels , No Activation, Width 1	$32 \rightarrow C$

3. For each wiring, the parings are determined. The channel values of the partners are collected and concatenated. A CNN (termed Non Local CNN) with 3 Layers act on this grid with $3C$ channels:

Layer 1	2D Convolution , 32 Channels, ReLU, Width 1	$1 \rightarrow 32$
Layer 2	2D Convolution , 32 Channels , ReLU, Width 1	$32 \rightarrow 32$
Layer 3	2D Convolution , C Channels , No Activation, Width 1	$32 \rightarrow C$

4. The Local CNN and the Non Local CNN outputs their own separate update grid. These are then independently added to the current grid and evolved.
5. The rest of the training procedure proceeds similar to the vanilla NCA.

This version of the Non Local NCA is discarded now and the reasons are discussed in Section 4.1.

2.2.2. v2

1. An identity filter, Sobel X filter, and Sobel Y filter are convolved over the grid ($C \rightarrow 3C$)
2. For each wiring, the parings are determined. Three additional channels are created which will contain the partnter's identity,

• which is a function of fraction of cells with wiring and the Manhattan Distance

• C is the number of channels

• If the cell deoesnt have a partner, zero vectors are concatenated, so the lenght of the vector of NC

Sobel X and Sobel Y values respectively. If a cell doesn't have a partner the values are kept at 0. (3C $\rightarrow$ 6C)

throughout the process

3. A CNN with 3 Layers act on this grid with 3C channels:

Layer 1	2D Convolution , 32 Channels, ReLU, Width 1	6C $\rightarrow$ 32
Layer 2	2D Convolution , 32 Channels , ReLU, Width 1	32 $\rightarrow$ 32
Layer 3	2D Convolution , C Channels , No Activation, Width 1	32 $\rightarrow$ C

4. The output of Layer 3 is the update given to the current grid.
5. The rest of the training procedure proceeds similar to the vanilla NCA.

2.3. Initialization

Each cell in the grid is initialized with a small random noise at 0. The weights of the network are also randomly chosen. The probability of a cell getting a long range partner is determined by a hyperparameter. A small fraction of cells are then converted to daughter A/ daughter B cells with random chance so that there is variability and symmetry-breaking in the initial state.

another hyperparameter

3. Experiments

3.1. Convergence Time Comparison

1. N_R number of ratios are randomly chosen between the range 0.35 and 0.75. Each N_R is associated with N_s number of seeds. The seeds are same across ratios.
2. The seed determines the model weights, the initial grid etc. so that both the vanilla NCA and Non Local NCA having almost everything same except the wiring topology.
3. A vanilla NCA model and a Non Local NCA Model is trained for every seed associated with every ratio.
4. N_{eval} number of fresh initial grids are then made and it is evolved with the learned rules till it reaches within tol. The maximum number of steps it can take is set as MAX_STEPS and the models that fail to converge before this are discarded.
5. The number of steps taken for convergence is noted down for both vanilla NCA and Non Local NCA. An average is taken across the evaluation grids and this number can be denoted as $t_{s_i}^{r_j}$

0 to 1 was not chosen because it was getting harder to train

for 1000 epochs

tol is set here as within 0.05 band of target ratio and less than 5% parent cells

Work has to be done to bring this to a minimum.

satisfying the tol condition

$$6. \quad \frac{1}{N_s + N_r} \sum_{s_i} \sum_{r_j} t_{s_i}^{r_j} \quad (8)$$

is the mean convergence time. This is shown as the point in the plot.

$$7. \quad \frac{\sigma_{\{s_i\} \cup \{r_j\}}}{\sqrt{N_s * N_r}} \quad (9)$$

is the mixed standard deviation of the readings and this is plotted as the error-bar

8. $(\sigma_{\{s_i\}})$ is the standard deviation of the readings across seed axis and this is plotted as a band.
9. Steps 4 to 8 are repeated again for grids of different sizes in N_{test}

3.1.1. Results

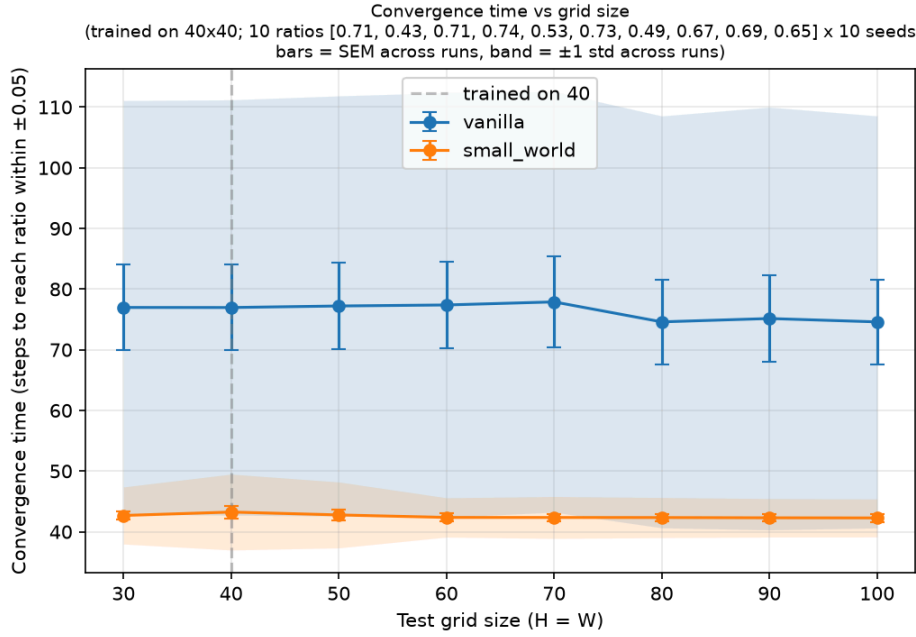


Figure 1: Convergence Time vs Grid Size for vanilla NCA and Non Local NCA with $p = 0.3$. The mean convergence time are the points, $\frac{\sigma}{\sqrt{N}}$ of the whole dataset is the error-bar, and σ across seeds is the band.

- From Figure 1 it is evident that Non Local NCA has a clear advantage in convergence time. Plus the convergence time across seeds are stable for Non Local NCA compared to vanilla NCA which are all over the place. Does this mean that the Non-Local NCA converged to a more general rule across seeds, wirings and initializations that is not affected by perturbation?

3.1.2. Controls

Since the channels 3-5 are always set to 0 for the vanilla NCA, a reason for the slower performance of vanilla NCA might be due to effectively working with a lower parameter count. There is a need to perform the experiment with the effective parameters of vanilla NCA either matching or exceeding that of its non local counterpart to strengthen the claim for non local connections. To address the effective parameters issue, two approaches were formed:

1. Increase the parameters in the CNN only for the vanilla NCA, so the effective parameters of both methods match.
2. Add itself as a partner for the cell, which is indirectly copying the information in channels 0-2 to channels 3-5 with the same probability for long range connections. This is a cleaner approach. The result of this approach is in Figure 2.

denoted as small-world

Parameters for Figure 1

N_r	10
N_s	10
N_{eval}	10

notice the small band for non local NCA compared to a very wide band for vanilla

because increasing the parameters changes the network, and it changes the

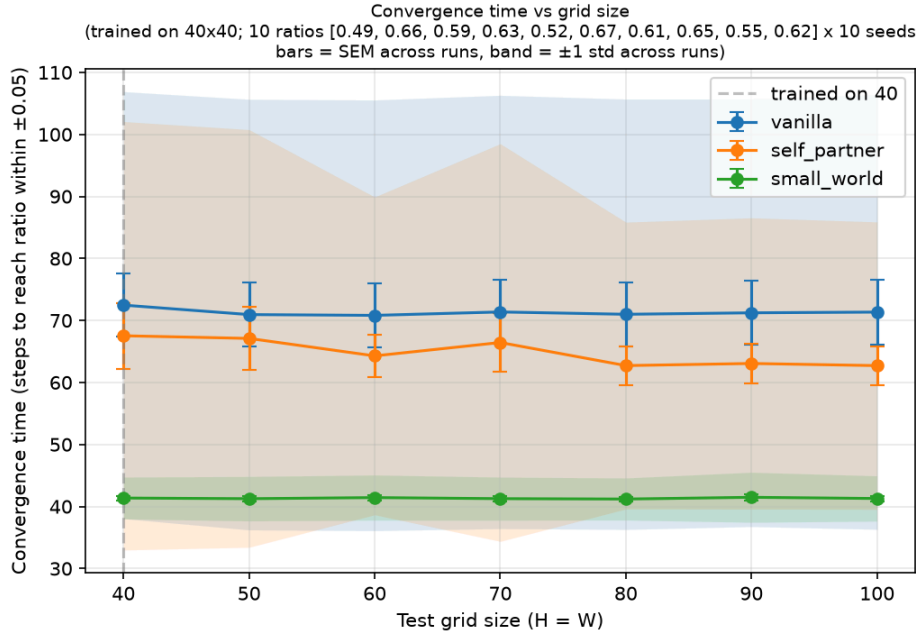


Figure 2: Convergence Time vs Grid Size for vanilla NCA and Non Local NCA and a self-partnered vanilla NCA with $p = 0.3$.

- Adding a self partner did not help with the stability or the convergence time, when compared with the vanilla approach. This means parameters are not causing the performance jump, and can be ruled out.

Another control experiment was performed with keeping 3 extra hidden channels active for the vanilla NCA, so that both vanilla and non-local NCA have 6 channels at all times. The results are shown in Figure 3.

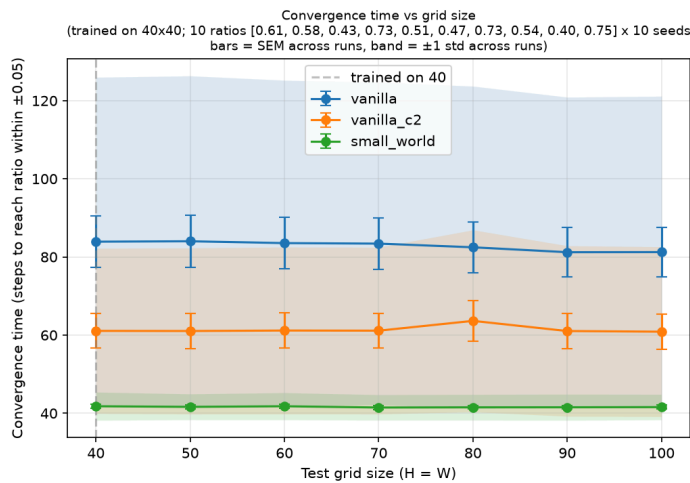


Figure 3: Convergence Time vs Grid Size for vanilla NCA with 3channels (blue), vanilla NCA with 6 channels Non (orange, denoted as vanilla_c2) and a non-local NCA and a self-partnered vanilla NCA with $p = 0.3$.

After discussions with people at lossfunk it was pointed out to me that the model was not learning anything and might be outputting the same output for any given input. A collapse check was performed comparing standard deviations between outputs of various inputs. The standard deviations tell that outputs are significantly different for different inputs. A visualisation of some of those outputs are given in Figure 4

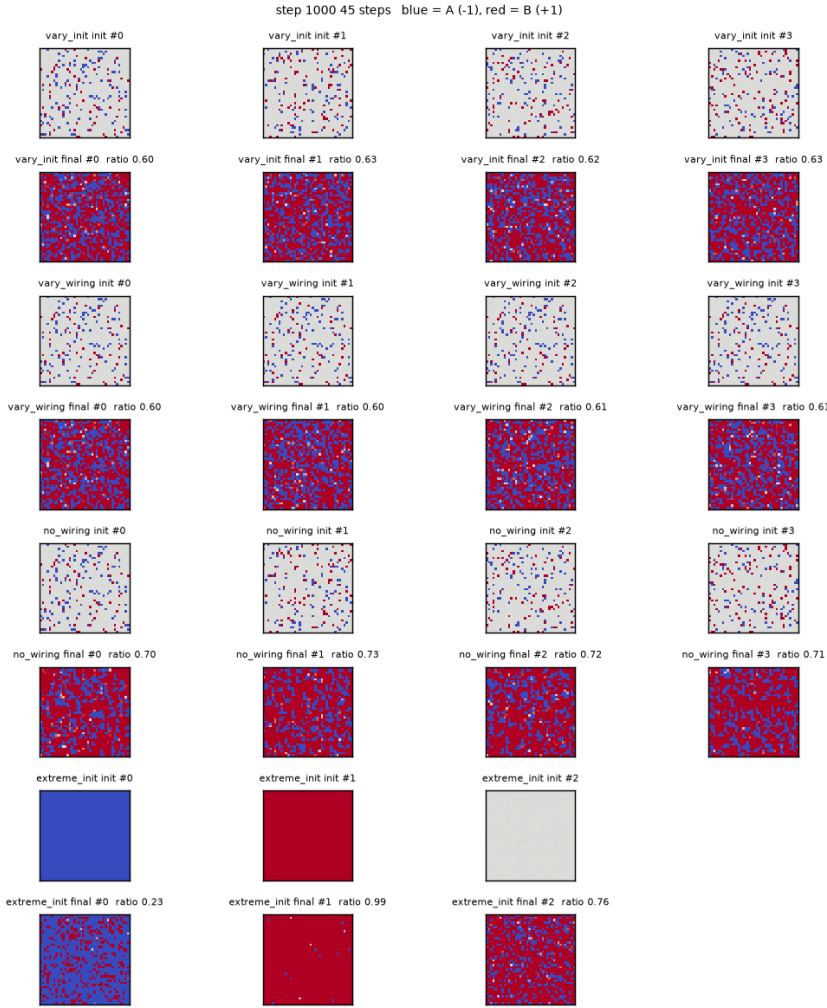


Figure 4: Initial grids and final grids plotted for different initial conditions. Row 0 and Row 1: Initial conditions with both initial grid and wiring varied. Row 2,3: Initial conditions with only wiring varied. Row 4,5: Initial conditions with only initial grid varied. Row 6,7: Extreme initial conditions with grids starting only with either Cell A, Cell B or parent cell.

What would happen if we vary the long range connection distance or probability?

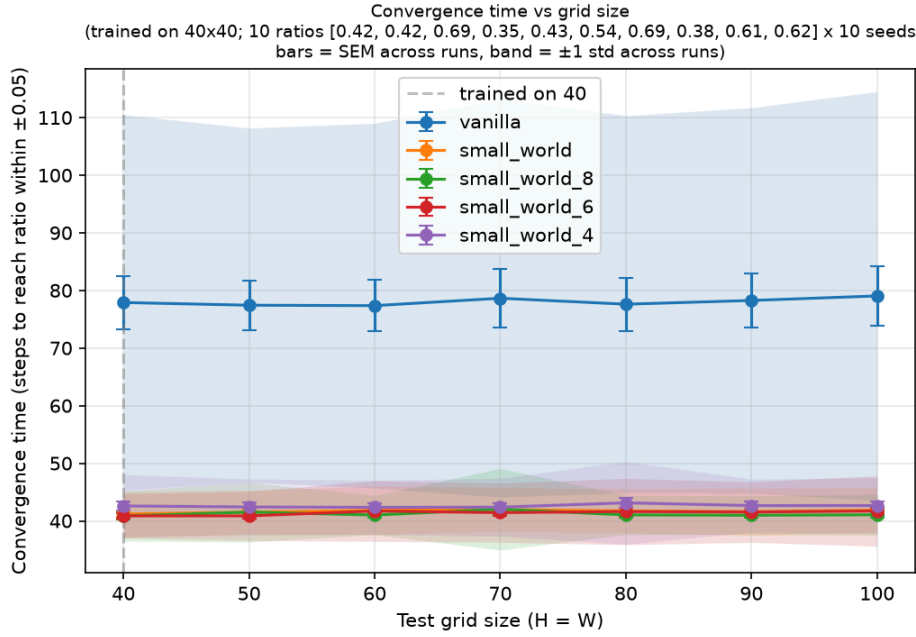


Figure 5: Convergence Time vs Grid Size for different Manhattan Distances i.e, the length of long range connections. small_world (orange) has a minimum Manhattan distance of 10 and no maximum. small_world_8 (green) has minimum Manhattan Distance 8 and maximum 10. small_world_6 has a minimum of 6 and maximum of 8; and so on till small_world_4.

A very surprising result from Figure 5, which shows the length of the long range connections do not influence the convergence time. A small world topology with a Manhattan Distance of 4 is showing the same performance of a topology with Manhattan Distance of 10.

The hunch is that probability of long-range connections is too much, and the distance effect might come into play at lower probabilities. Experiments sweeping the probability of a long range connection parameter are to be performed next.

3.1.3. Issues

1. Convergence time is staying same along grid size, which is contrary to the hypothesis. If the nature of the problem was to leverage global information, it should be the case that as the grid size increases, the cells should spend longer time to assess it's global state and therefore we should see a proportional relationship, which is not the case here. This maybe a huge hint that the network is learning a rule which can carry out the task only with local information, so there might be need to revisit the task definition.

4. Appendix

4.1. Why v1 was discarded

v1 had two networks, one for local and one for non local. In the MLP for the non local case, the values from the long range connections were concatenated and passed as input to the CNN. The concatenation process means that the connection information is lost in the process, which was the reason for introducing v2.

♥ since there is no way for one to know which part of the input came from which connection

Bibliography

- [1] A. Mordvintsev, E. Randazzo, E. Niklasson, and M. Levin, “Growing Neural Cellular Automata,” *Distill*, 2020, doi: 10.23915/distill.00023^o.

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$\heartsuit$ denoted as small-world

$\heartsuit$ Parameters for Figure 1

N_r	10
N_s	10
N_{eval}	10

$\heartsuit$ denoted as small-world

$\heartsuit$ denoted as small-world